

JAYDEN METCALFE

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TECHNICAL SKILLS:

Languages: Python, TypeScript, JavaScript, Java, C, Bash, Clojure

Frontend: React, Angular, HTML, CSS, React Testing Library, Vitest

Backend: Node.js, Express.js, FastAPI, OAuth 2.0, Jest

Databases: PostgreSQL

Cloud & DevOps: AWS, Docker, Docker Compose, REST APIs

Machine Learning & Data: scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib

Tools: Git, Arduino, Linux

WORK EXPERIENCE:

Full Stack Software Developer & U16 Alpine Ski Racing Coach

November 2024 - present

Whistler Mountain Ski Club, Whistler BC

- Engineered a full-stack analytics web application using React, TypeScript, Node.js, Express.js, PostgreSQL, and Python to track and analyze athletes' training data
- Implemented secure OAuth 2.0 authentication and role-based access control for coaches, athletes, and parents
- Architected and optimized RESTful APIs and SQL queries to support filtering and comparative analytics of hundreds of data points, improving query performance and enabling data-driven adjustments to team training programs
- Containerized using Docker and Docker Compose to separate frontend, backend, and database services, improving scalability and maintainability, then deployed the application as an Amazon Web Services (AWS) EC2 instance
- Built unit and integration test suites using Jest, Vitest, and React Testing Library, improving application reliability
- Collaborated with users to gather feedback and rapidly deliver feature updates in an iterative development process, often achieving same-day turnaround
- Analyzed data to design and lead targeted training sessions for 40+ athletes while coordinating with a team of 6 coaches to optimize program structure and outcomes

Reinforcement Learning Researcher

August 2023 - May 2024

The Britt Lab, Montréal QC

- Automated data processing and reporting using Python and Excel, generating visualizations from thousands of data points to support behavioural research analysis
- Implemented Q-learning models to simulate and analyze decision-making behaviour
- Debugged and extended Arduino code to scale for multiple experimental stages, each with variables that required fine tuning to maximize the collection of usable data points

Product Research, Design, and Development Intern

May - August 2023

Interac Corp., Toronto ON

- Prototyped RESTful API endpoints and a PostgreSQL database schema for an escrow-based e-transfer solution to enhance transaction security and reduce fraud risk, which was demonstrated to senior leadership
- Collaborated with cross-functional teams to build and update frontend features of an emulator using TypeScript and Angular within an MVC architecture, implementing UI/UX designs from Figma for client-specific product demos
- Refactored legacy code to improve maintainability, scalability, and integration with new product features

PERSONAL PROJECTS:

FIS Race Points Calculator

- Developed a full-stack web application using Python, Node.js, and REST APIs to calculate ski race scores in real time, reducing processing time from several hours to seconds
- Integrated external APIs to process live timing data and automate scoring calculations
- Built a responsive frontend using HTML, CSS, and JavaScript to display real-time results

EDUCATION:

Bachelor of Arts and Science, Cognitive Science

August 2020 - April 2024

McGill University, Montréal QC, Canada

- Neuroscience stream, Minor in Computer Science

Relevant Coursework: Algorithms & Data Structures, Python, Java & OOP, Probability, Software Systems, Computational Linguistics, Discrete Structures, Applied Machine Learning, Natural Language to Data Science, Linear Algebra, Deductive Logic